

## Hints &amp; Solutions

#MathBoleTohMathonGo

## Q1 (2) - Sine Rule and Its Application

In  $\Delta PQR$ 

$$\frac{\sin P}{p} = \frac{\sin Q}{q} = \frac{\sin R}{r} = \lambda \text{ (using sine rule)}$$

$$\text{Given, } \sin P \sin Q = \frac{pq}{r^2}$$

$$\Rightarrow \sin P \sin Q = \frac{\sin P \sin Q}{\sin^2 R}$$

$$\Rightarrow \sin^2 R = 1$$

$$\sin R = 1 = \sin \frac{\pi}{2}$$

$$\therefore R = \frac{\pi}{2} = 90^\circ$$

## Q2 (1) - Sine Rule and Its Application

$$\text{In } \Delta ABC, \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} = \lambda \text{ (let)}$$

$$\text{Now, } a^2 (\cos^2 B - \cos^2 C) + b^2 (\cos^2 C - \cos^2 A) + c^2 (\cos^2 A - \cos^2 B)$$

$$= a^2 (\sin^2 C - \sin^2 B) + b^2 (\sin^2 A - \sin^2 C) + c^2 (\sin^2 B - \sin^2 A)$$

$$= a^2 (\lambda^2 c^2 - \lambda^2 b^2) + b^2 (\lambda^2 a^2 - \lambda^2 c^2) + c^2 (\lambda^2 b^2 - \lambda^2 a^2)$$

$$= \lambda^2 (a^2 c^2 - a^2 b^2 + a^2 b^2 - b^2 c^2 + b^2 c^2 - a^2 c^2)$$

$$= 0$$

## Q3 (1) - Sine Rule and Its Application

Let in  $\Delta ABC$ ,

$$A : B : C = 4 : 1 : 1$$

$$\text{and } A = 4x, B = x, C = x$$

$$\text{We know that, } A + B + C = 180^\circ$$

$$4x + x + x = 180^\circ$$

$$x = 30^\circ$$

$$\therefore A = 120^\circ, B = 30^\circ, C = 30^\circ$$

$$\sin A = \frac{\sqrt{3}}{2}, \sin B = \frac{1}{2}, \sin C = \frac{1}{2}$$

Now, using sine rule, we get

Hints & Solutions

#MathBoleTohMathonGo

$$a = \lambda \sin A, b = \lambda \sin B, c = \lambda \sin C$$

$$a = \frac{\lambda\sqrt{3}}{2}, b = \frac{\lambda}{2}, c = \frac{\lambda}{2}$$

$$\text{Perimeter of } \triangle ABC = a + b + c = \frac{\lambda(\sqrt{3} + 2)}{2}$$

$$\text{and largest side} = \frac{\lambda\sqrt{3}}{2}$$

$$\therefore \text{Required ratio} = \frac{\lambda\sqrt{3}}{2} : \frac{\lambda(\sqrt{3} + 2)}{2}$$

$$= \sqrt{3} : (2 + \sqrt{3})$$

Q4 (3) - Sine Rule and Its Application

$$\text{In } \triangle ABC, \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = \lambda \text{ (let)}$$

$$\text{We have, } \frac{a^2 - b^2}{a^2 + b^2} = \frac{\sin(A - B)}{\sin(A + B)}$$

$$\frac{\lambda^2 \sin^2 A - \lambda^2 \sin^2 B}{\lambda^2 \sin^2 A + \lambda^2 \sin^2 B} = \frac{\sin(A - B)}{\sin(A + B)}$$

$$\sin^2(A + B) \sin(A - B) = \sin(A - B) (\sin^2 A + \sin^2 B)$$

$$\sin^2(\pi - C) \sin(A - B) - \sin(A - B) (\sin^2 A + \sin^2 B) = 0$$

$$\sin(A - B) [\sin^2 C - (\sin^2 A + \sin^2 B)] = 0$$

$$\text{Either } \sin(A - B) = 0$$

$$\Rightarrow A - B = 0$$

$$\Rightarrow A = B$$

$$\text{or } \sin^2 C - (\sin^2 A + \sin^2 B) = 0$$

$$\sin^2 C = \sin^2 A + \sin^2 B$$

$$c^2 = a^2 + b^2$$

$\therefore \triangle ABC$  is a right angled or isosceles.

Q5 (4) - Cosine Rule and Projection Formulae

$$\therefore \frac{\cos A}{b \cos C + c \cos B} + \frac{\cos B}{c \cos A + a \cos C} + \frac{\cos C}{a \cos B + b \cos A}$$

$$= \frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} \quad (\text{using projection formula})$$

$$= \frac{b^2 + c^2 - a^2}{2abc} + \frac{c^2 + a^2 - b^2}{2abc} + \frac{a^2 + b^2 - c^2}{2abc} \quad (\text{using cosine rule})$$

$$= \frac{b^2 + c^2 - a^2 + c^2 + a^2 - b^2 + a^2 + b^2 - c^2}{2abc}$$

$$= \frac{a^2 + b^2 + c^2}{2abc}$$

Hints & Solutions

#MathBoleTohMathonGo

Now, we have  $\frac{a^2+b^2+c^2}{\lambda abc} = \frac{a^2+b^2+c^2}{2abc}$

$$\lambda = 2$$

Q6 (1) - Cosine Rule and Projection Formulae

We have,

$$(a + b + c)(a - b + c) = 3ac$$

$$(a + c)^2 - b^2 = 3ac$$

$$(a^2 + c^2 - b^2) + 2ac = 3ac$$

$$\frac{a^2 + c^2 - b^2}{2ac} = \frac{ac}{2ac}$$

$$\frac{a^2 + c^2 - b^2}{2ac} = \frac{1}{2}$$

$$\Rightarrow \cos B = \cos 60^\circ$$

$$\therefore \angle B = 60^\circ$$

Q7 (3) - Cosine Rule and Projection Formulae

We have,

$$\frac{2 \cos A}{a} + \frac{\cos B}{b} + \frac{2 \cos C}{c} = \frac{a}{bc} + \frac{b}{ca}$$

$$\frac{2(b^2+c^2-a^2)}{2abc} + \frac{c^2+a^2-b^2}{2abc} + \frac{2(a^2+b^2-c^2)}{2abc} = \frac{a^2+b^2}{abc}$$

$$2b^2 + 2c^2 - 2a^2 + c^2 + a^2 - b^2 + 2a^2 + 2b^2 - 2c^2 = 2a^2 + 2b^2$$

$$b^2 + c^2 = a^2$$

Hence, the value of angle A is  $90^\circ$

Q8 (1) - Cosine Rule and Projection Formulae

We know that,

$$A + B + C = 180^\circ$$

$$2B + B = 180^\circ \quad (\because 2B = A + C, \text{ given})$$

$$B = 60^\circ$$

Then,  $\cos B = \frac{c^2+a^2-b^2}{2ca}$  (using cosine rule)

$$\cos 60^\circ = \frac{c^2 + a^2 - b^2}{2ca}$$

$$\frac{1}{2} = \frac{c^2 + a^2 - b^2}{2ca}$$

$$b^2 = c^2 + a^2 - ca \quad \dots(i)$$

Now,  $\frac{a+c}{\sqrt{a^2+c^2-ac}} = \frac{a+c}{b}$  [from Eq. (i)]

$$= \frac{\sin A + \sin C}{\sin B} \quad (\text{using sine rule})$$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

$$\begin{aligned}
 &= \frac{2 \sin\left(\frac{A+C}{2}\right) \cos\left(\frac{A-C}{2}\right)}{2 \sin \frac{B}{2} \cos \frac{B}{2}} \\
 &= \frac{\cos \frac{B}{2} \cos\left(\frac{A-C}{2}\right)}{\sin \frac{B}{2} \cos \frac{B}{2}} \left( \because \frac{A+B+C=180^\circ}{2} \right) \\
 &= \frac{\cos\left(\frac{A-C}{2}\right)}{\sin 30^\circ} = 2 \cos\left(\frac{A-C}{2}\right) \quad (\because B = 60^\circ)
 \end{aligned}$$

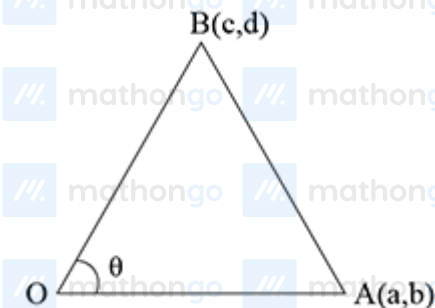
Q9 (2) - Cosine Rule and Projection Formulae

We have, in  $\triangle ABC$ ,  $b = c$ , then

$$\angle B = \angle C$$

$$\therefore \angle B = \angle C = \frac{160^\circ}{2} = 80^\circ$$

$$(\because \angle A = 20^\circ, \text{ given})$$



Now, using sine rule, we get

$$\frac{a}{\sin 20^\circ} = \frac{b}{\sin 80^\circ} = \frac{c}{\sin 80^\circ}$$

$$\text{If } \frac{a}{\sin 20^\circ} = \frac{b}{\sin 80^\circ} = \frac{b}{\sin(90^\circ - 10^\circ)}$$

$$\Rightarrow \frac{a}{\sin 20^\circ} = \frac{b}{\cos 10^\circ}$$

$$\Rightarrow a = \frac{b \sin 20^\circ}{\cos 10^\circ} = \frac{2b \sin 10^\circ \cos 10^\circ}{\cos 10^\circ}$$

$$\Rightarrow a = 2b \sin 10^\circ \quad (\text{i})$$

$$\text{Thus, } a^3 + b^3 = (2b \sin 10^\circ)^3 + b^3$$

Q10 (1) - Napier Analogy and Half Angle Formulae

$$1 - \tan \frac{B}{2} \cdot \tan \frac{C}{2} = 1 - \sqrt{\frac{(s-a)(s-c)}{s(s-b)}} \cdot \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$$

$$= 1 - \frac{s-a}{s} = \frac{s-s+a}{s}$$

$$= \frac{a}{s} = \frac{a}{\frac{a+b+c}{2}} = \frac{2a}{a+b+c}$$

Q11 (4) - Napier Analogy and Half Angle Formulae

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

$$\begin{aligned} \text{We have, } \sin \frac{B}{2} \sin \frac{C}{2} &= \frac{b+c-a}{a} \\ \Rightarrow 2 \sin \frac{B}{2} \sin \frac{C}{2} &= \frac{2(b+c-a)}{a} \\ \Rightarrow \cos \left( \frac{B}{2} - \frac{C}{2} \right) - \cos \left( \frac{B}{2} + \frac{C}{2} \right) &= \frac{2(b+c-a)}{a} \\ \Rightarrow \cos \frac{B}{2} \cos \frac{C}{2} + \sin \frac{B}{2} \sin \frac{C}{2} - \cos \left( \frac{\pi}{2} - \frac{A}{2} \right) &= \frac{2(b+c-a)}{a} \\ \Rightarrow \sqrt{\frac{s(s-b)}{ac}} \cdot \sqrt{\frac{s(s-c)}{ab}} + \sqrt{\frac{(s-a)(s-c)}{ac}} \cdot \sqrt{\frac{(s-a)(s-b)}{ab}} - \sin \frac{A}{2} &= \frac{2(b+c-a)}{a} \\ \Rightarrow \frac{s}{a} \cdot \sqrt{\frac{(s-b)(s-c)}{bc}} + \frac{s-a}{a} \cdot \sqrt{\frac{(s-b)(s-c)}{bc}} - \sqrt{\frac{(s-b)(s-c)}{bc}} &= \frac{2(b+c-a)}{a} \\ \Rightarrow \sqrt{\frac{(s-b)(s-c)}{bc}} \left[ \frac{s}{a} + \frac{s-a}{a} - 1 \right] &= \frac{2(b+c-a)}{a} \\ \Rightarrow \sin \frac{A}{2} \left[ \frac{s+s-a-a}{a} \right] &= \frac{2(b+c-a)}{a} \\ \Rightarrow \sin \frac{A}{2} \left[ \frac{2s-2a}{a} \right] &= \frac{2(b+c-a)}{a} \\ \Rightarrow \sin \frac{A}{2} \left[ \frac{a+b+c-2a}{a} \right] &= \frac{2(b+c-a)}{a} \\ \Rightarrow \sin \frac{A}{2} \left[ \frac{b+c-a}{a} \right] &= \frac{2(b+c-a)}{a} \\ \therefore \sin \frac{A}{2} &= 2, \text{ which is not possible.} \end{aligned}$$

Q12 (1) - Napier Analogy and Half Angle Formulae

$$\begin{aligned} \text{We have, } \sin \frac{A}{2} \sin \frac{B}{2} &= \sin \frac{C}{2} \\ \sqrt{\frac{(s-b)(s-c)}{bc}} \cdot \sqrt{\frac{(s-a)(s-c)}{ac}} b &= \sqrt{\frac{(s-a)(s-b)}{ab}} \\ \Rightarrow \frac{s-c}{c} &= 1 \\ \Rightarrow s - c &= c \\ \therefore s &= 2c \end{aligned}$$

Q13 (2) - Napier Analogy and Half Angle Formulae

$$\begin{aligned} \text{We have, } \cot \frac{A}{2} \cot \frac{B}{2} &= c \\ \sqrt{\frac{s(s-a)}{(s-b)(s-c)}} \cdot \sqrt{\frac{s(s-b)}{(s-a)(s-c)}} s - c &= c \\ \frac{s}{s-c} &= c \\ \frac{1}{s-c} &= \frac{c}{s} \dots (i) \\ \text{Similarly, } \cot \frac{B}{2} \cot \frac{C}{2} &= a \\ \Rightarrow \frac{1}{s-a} &= \frac{a}{s} \end{aligned}$$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

$$\text{and } \cot \frac{C}{2} \cot \frac{A}{2} = b$$

$$\Rightarrow \frac{1}{s-b} = \frac{b}{s}$$

$$\text{Now, } \frac{1}{s-a} + \frac{1}{s-b} + \frac{1}{s-c} = \frac{a}{s} + \frac{b}{s} + \frac{c}{s} \text{ [from Eqs. (i), (ii) and (iii)]}$$

$$= \frac{a+b+c}{s}$$

$$= \frac{2s}{s} = 2$$

Q14 (2) - Napier Analogy and Half Angle Formulae

$$\text{We have, } \tan \frac{A}{2} = \frac{5}{6}, \tan \frac{C}{2} = \frac{2}{5}$$

$$\text{Then, } \cot \frac{A}{2} = \frac{6}{5}, \cot \frac{C}{2} = \frac{5}{2}$$

$$\therefore \cot \frac{A}{2} \cot \frac{C}{2} = 3$$

$$\Rightarrow \sqrt{\frac{s(s-a)}{(s-b)(s-c)}} \cdot \sqrt{\frac{s(s-c)}{(s-a)(s-b)}} = 3$$

$$\Rightarrow \frac{s}{s-b} = 3$$

$$\Rightarrow s = 3s - 3b$$

$$\Rightarrow 2s - 3b = 0$$

$$\Rightarrow a + c - 2b = 0$$

$$\therefore 2b = a + c$$

Hence,  $a, b, c$  are in AP.

Q15 (2) - Area of Triangle

We know that,

$$\Delta = \frac{1}{2} \cdot \frac{a^2 \sin B \sin C}{\sin(B+C)}$$

$$= \frac{1}{2} \cdot \frac{(4)^2 \cdot \sin 30^\circ \sin 45^\circ}{\sin 75^\circ}$$

$$= \frac{8 \cdot \frac{1}{2} \cdot \frac{1}{\sqrt{2}}}{\frac{\sqrt{3}+1}{2\sqrt{2}}} = \frac{8}{\sqrt{3}+1}$$

$$= \frac{8(\sqrt{3}-1)}{3-1}$$

$$= 4(\sqrt{3}-1)\text{cm}^2$$

Q16 (1) - Area of Triangle

#MathBoleTohMathonGo

Hints & Solutions

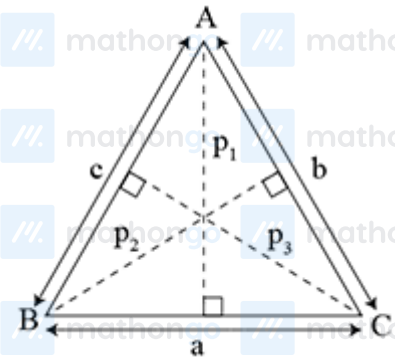
#MathBoleTohMathonGo

We know that,

$$\Delta = \frac{1}{2} \times \text{Base} \times \text{Corresponding altitude}$$

$$\Delta = \frac{1}{2} \times a \times p_1$$

$$\therefore p_1 = \frac{2\Delta}{a}$$



$$\text{Similarly, } p_2 = \frac{2\Delta}{b}$$

$$\text{and } p_3 = \frac{2\Delta}{c}$$

It is given that  $p_1, p_2, p_3$  are in HP.

$$\frac{2\Delta}{a}, \frac{2\Delta}{b}, \frac{2\Delta}{c} \text{ are in HP.}$$

On dividing each by  $2\Delta$ , we get

$$\frac{1}{a}, \frac{1}{b}, \frac{1}{c} \text{ are in HP.}$$

Thus,  $a, b, c$  are in AP.

$\therefore \sin A, \sin B, \sin C$  are also in AP. (using formula)

**Q17 (1) - Area of Triangle**

Let  $a, b, c$  be the sides of a triangle, then

$$\Delta = 20\sqrt{3} \text{ cm}^2$$

We know that,

$$\Delta = \frac{1}{2} ab \sin C$$

$$\Rightarrow 20\sqrt{3} = \frac{1}{2} ab \sin 60^\circ$$

$$20\sqrt{3} = \frac{1}{2} ab \frac{\sqrt{3}}{2}$$

$$\Rightarrow ab = 80 \quad \dots (i)$$

We have,

#MathBoleTohMathonGo



Hints & Solutions

#MathBoleTohMathonGo

$$a + b + c = 20$$

$$a + b = 20 - c$$

On squaring,

$$(a + b)^2 = (20 - c)^2 \quad \dots(ii)$$

$$\text{Now, } a^2 + b^2 = (a + b)^2 - 2ab$$

$$= (20 - c)^2 - 2 \times 80 \text{ [from Eqs. (i) and (ii)]}$$

$$= 400 + c^2 - 40c - 160$$

$$\Rightarrow a^2 + b^2 = c^2 - 40c + 240 \dots(iii)$$

Using cosine rule, we get

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\Rightarrow \cos 60^\circ = \frac{c^2 - 40c + 240 - c^2}{2 \times 80} \text{ [from Eqs. (i) and (iii)]}$$

$$\Rightarrow \frac{1}{2} = \frac{240 - 40c}{2 \times 80}$$

$$\Rightarrow 240 - 40c = 80$$

$$\Rightarrow 40c = 240 - 80$$

$$\therefore C = \frac{160}{40} = 4 \text{ cm}$$

Q18 (2) - Area of Triangle

We have.

$$a^2 + b^2 = 1999c^2 \dots(i)$$

$$\text{Now, } 999 \left( \frac{\cot A + \cot B}{\cot C} \right) = 999 \left[ \frac{\frac{\cos A}{\sin A} + \frac{\cos B}{\sin B}}{\frac{\cos C}{\sin C}} \right]$$

$$= 999 \left[ \frac{\frac{bc \cos A}{2\Delta} + \frac{ac \cos B}{2\Delta}}{\frac{ab \cos C}{2\Delta}} \right] \text{ (using } \Delta = \frac{1}{2}bc \sin A \text{)}$$

$$= 999 \left[ \frac{c(b \cos A + a \cos B)}{ab \cos C} \right]$$

$$= 999 \left[ \frac{2 \cdot c \cdot c}{2ab \cdot \frac{a^2 + b^2 - c^2}{2ab}} \right]$$

$$\text{(using projection formula and cosine rule)} = 999 \times \left( \frac{2c^2}{1999c^2 - c^2} \right) \text{ [from Eq.(i)]}$$

$$= \frac{1998c^2}{1998c^2} = 1$$

Q19 (3) - Area of Triangle

We know that,  $AM \geq GM$

$$\text{Then, } \frac{(s-a) + (s-b) + (s-c)}{3} \geq [(s-a)(s-b)(s-c)]^{1/3}$$

#MathBoleTohMathonGo



Hints & Solutions

#MathBoleTohMathonGo

$$\Rightarrow \frac{3s-(a+b+c)}{3} \geq \left[ \frac{s(s-a)(s-b)(s-c)}{s} \right]^{1/3}$$

$$\Rightarrow \left( \frac{s}{3} \right)^3 \geq \frac{\Delta^2}{s} \quad \left( \because a+b+c=2s \text{ and Hero's formula} \right)$$

$$\Rightarrow \Delta^2 \leq \frac{s^4}{(3)^3} = \frac{s^4}{27}$$

$$\Rightarrow \Delta \leq \frac{s^2}{3\sqrt{3}}$$

But given that  $\Delta \leq \frac{s^2}{\lambda\sqrt{3}}$ , then comparing, we get

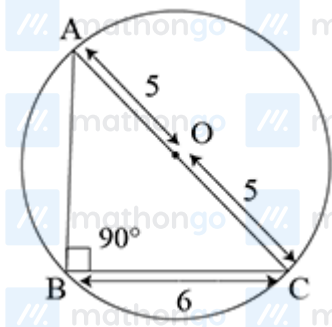
$$\lambda = 3$$

Q20 (1) - Properties of Triangles

In right angled  $\triangle ABC$

$$\angle ABC = 90^\circ, a = 6, R = 5$$

$$\text{Then, } AC = OA + OC = 2OC \quad (\because OA = OC = R) \\ = 10$$



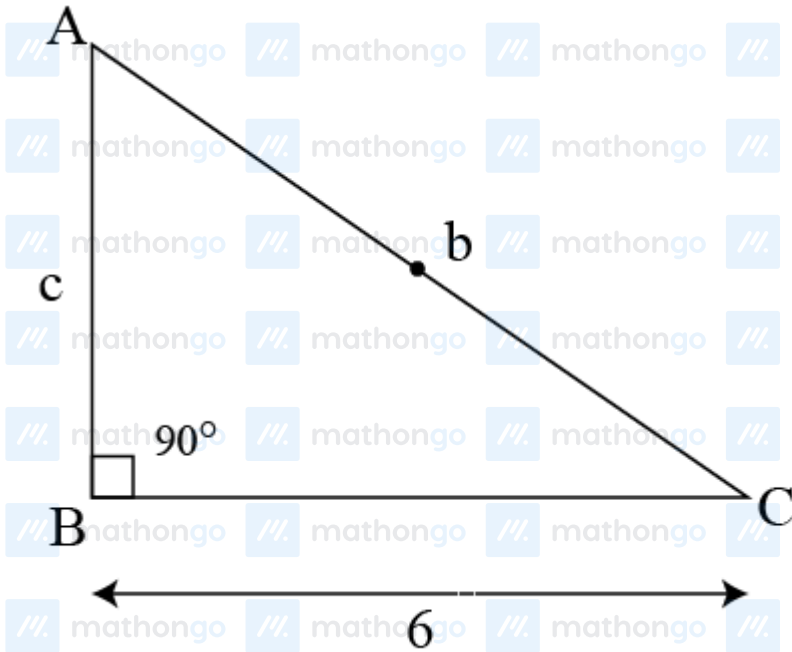
Now, using Pythagoras theorem,

$$AB^2 = AC^2 - BC^2 = (10)^2 - (6)^2 \\ = 100 - 36 = 64 \\ AB = 8$$

## Hints &amp; Solutions

#MathBoleTohMathonGo

$$\begin{aligned}\therefore \text{Area of } \triangle ABC &= \frac{1}{2} \cdot AB \cdot BC \\ &= \frac{1}{2} \times 8 \times 6 = 24 \text{ cm}^2\end{aligned}$$



## Q21 (4) - Properties of Triangles

In  $\triangle ABC$ , let  $a = 3$ ,  $b = 4$ ,  $c = 5$ , then

$$s = \frac{a+b+c}{2} = \frac{3+4+5}{2}$$

$$s = 6$$

$$\text{and } \Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{6(6-3)(6-4)(6-5)}$$

$$\Delta = 6$$

We know that,

$$r = \frac{\Delta}{s} = \frac{6}{6} = 1$$

$\therefore$  Inradius  $(r) = 1$

## Q22 (4) - Properties of Triangles

We have, in  $\triangle ABC$ ,  $a = 5$ ,  $b = 12$ ,  $\angle C = 90^\circ$ ,

then using the cosine rule,

## Hints &amp; Solutions

#MathBoleTohMathonGo

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\cos 90^\circ = \frac{5^2 + 12^2 - c^2}{2 \times 5 \times 12}$$

$$0 = 169 - c^2 \quad (\because \cos 90^\circ = 0)$$

$$c = 13$$

Now, we know that,

$$\frac{a}{2 \sin A} = \frac{b}{2 \sin B} = \frac{c}{2 \sin C} = R$$

$$\therefore R = \frac{C}{2 \sin C} = \frac{13}{2 \cdot \sin 90^\circ}$$

$$\therefore R = \frac{13}{2}$$

## Q23 (3) - Properties of Triangles

We know that,

$$\frac{a}{2 \sin A} = \frac{b}{2 \sin B} = \frac{C}{2 \sin C} = R$$

$$\text{Then, } \sin A = \frac{a}{2R}, \sin B = \frac{b}{2R}, \sin C = \frac{c}{2R}$$

$$\text{Now, } 2R^2 \sin A \sin B \sin C = 2R^2 \cdot \frac{a}{2R} \cdot \frac{b}{2R} \cdot \frac{c}{2R}$$

$$= \frac{abc}{4R} = \Delta \quad \left( \because R = \frac{abc}{4\Delta} \right)$$

## Q24 (3) - Properties of Triangles

We know that,

$$\frac{a}{2 \sin A} = \frac{b}{2 \sin B} = \frac{C}{2 \sin C} = R$$

$$\Rightarrow \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R \dots (i)$$

Now,  $a \cot A + b \cot B + c \cot C$ 

$$= \frac{a}{\sin A} \cdot \cos A + \frac{b}{\sin B} \cdot \cos B + \frac{C}{\sin C} \cdot \cos C$$

$$= 2R(\cos A + \cos B + \cos C) \quad [\text{from Eq. (i)}]$$

$$= 2R \left( 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \right)$$

$$= 2R \left( 1 + \frac{r}{R} \right) \quad \left( \because r = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \right)$$

$$= 2(R + r)$$

## Q25 (1) - Properties of Triangles

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

We have,  $p, q, r$  be the sides of a  $\Delta PQR$ , then

$$R = \frac{pqr}{4\Delta} \text{ and } r = \frac{\Delta}{s}$$

$$\text{Now, } \frac{R}{r} = \frac{pqr}{4\Delta} \cdot \frac{s}{\Delta} = \frac{pqr}{4\Delta^2} \cdot s$$

$$\text{Let } p = 4\lambda, q = 5\lambda, r = 6\lambda$$

$$\text{Then, } s = \frac{p+q+r}{2} = \frac{4\lambda+5\lambda+6\lambda}{2}$$

$$s = \frac{15\lambda}{2}$$

and using Hero's formula

$$\begin{aligned} \Delta^2 &= s(s-p)(s-q)(s-r) \\ &= \frac{15\lambda}{2} \left( \frac{15\lambda}{2} - 4\lambda \right) \left( \frac{15\lambda}{2} - 5\lambda \right) \left( \frac{15\lambda}{2} - 6\lambda \right) \\ &= \frac{15\lambda}{2} \cdot \frac{7\lambda}{2} \cdot \frac{5\lambda}{2} \cdot \frac{3\lambda}{2} \end{aligned}$$

$$\Delta^2 = \frac{225 \times 7\lambda^4}{16}$$

$$\text{From Eq. (i), } \frac{R}{r} = \frac{4\lambda \cdot 5\lambda \cdot 6\lambda}{\frac{4 \times 225 \times 7\lambda^4}{16} \cdot \frac{15\lambda}{2}} = \frac{16}{7}$$

$$\therefore R : r = 16 : 7$$

Q26 (1) - Properties of Triangles

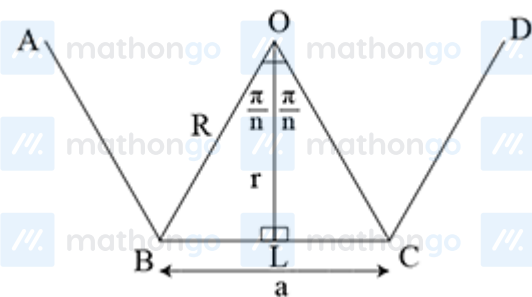
In an  $n$ -sided regular polygon

$$\angle BOC = \frac{2\pi}{n}$$

$$BL = LC = \frac{1}{2}BC$$

$$\text{and } \angle BOL = \angle COL = \frac{1}{2}\angle BOC$$

$$\therefore \angle BOL = \angle COL = \frac{\pi}{n}$$



Let circumradius  $(OB) = R$  and inradius  $(OL) = r$

In  $\Delta OLB$ ,

**#MathBoleTohMathonGo**